

Advanced Prompting Module - Assessment Design

Project Part 2

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Part 1: Initial Assessment Design

Instructional Approach

My educational philosophy centers on experiential learning and practical application, particularly in technical fields like AI prompting. I believe that advanced concepts are best learned through hands-on practice with real-world applications. My instructional approach emphasizes:

1. Interactive demonstrations: Live examples of advanced prompting techniques
2. Hands-on practice: Students craft and refine prompts using various AI tools
3. Peer review: Students evaluate and provide feedback on each other's prompts
4. Real-world application: Focus on prompting scenarios relevant to legal/business contexts
5. Reflective practice: Students analyze prompt effectiveness and iterate on approaches

Context

- Learning context: Online asynchronous course, Week 5 module
- Learners: 25-30 adult professionals enrolled in Master of Legal Studies program
- Prior knowledge: Basic understanding of AI and simple prompting techniques
- Learning setting: Part of "AI and the Law for MLS" course
- Time-frame: One week (Week 5 of 13-week semester)

Module Description

This module, "Advanced Prompting Techniques," builds upon basic prompting skills learned earlier in the course. Students explore sophisticated methods to enhance AI output quality and relevance in legal and professional contexts.

Major Learning Objectives:

1. Understand advanced prompting techniques (Chain of Thought, Few-Shot Learning, PCRO)
2. Apply these techniques to improve AI responses in legal/business scenarios
3. Evaluate the effectiveness of different prompting strategies
4. Develop ability to iteratively refine prompts for optimal results
5. Understand ethical considerations in advanced AI prompting

Assessment Plan

Formative Assessments (30%):

1. Daily prompt challenges (20%) • Purpose: Practice applying advanced prompting techniques • Format: Short, scenario-based prompting exercises • Frequency: Daily throughout the week • AI Usage: Students

- must document which AI tools they use and include their prompts • Time Commitment: 20-30 minutes per challenge
2. Peer feedback on prompts (10%) • Purpose: Develop critical evaluation skills • Format: Structured feedback on classmates' prompts • Frequency: 2-3 times during week • Time Commitment: 15-20 minutes per review

Summative Assessments (70%):

1. Advanced Prompting Quiz (30%) • Format: Multiple choice, short answer, and scenario-based questions • AI Usage: Not permitted for quiz portion • Time: 90 minutes • Scheduling: Flexible window within 24 hours to accommodate working professionals
2. Capstone Prompting Project (40%) • Format: Design a series of prompts for a complex legal/business scenario • AI Tools: Students must use at least two different AI platforms • Documentation: Required logging of all AI interactions • Time Frame: One week to complete • Presentation: Brief video presentation (5 minutes)

Table of Specifications

Content Area	Cognitive Level	Weight
Understanding Advanced Prompting Techniques	Knowledge (10%), Comprehension (10%)	20%
Applying Advanced Techniques	Application (20%), Analysis (15%)	35%
Evaluating Prompting Strategies	Analysis (15%), Evaluation (10%)	25%
Ethical Considerations	Comprehension (10%), Evaluation (10%)	20%

Part 2: Detailed Assessment Implementation

Assessment Design

Advanced Prompting Quiz (30%)

Multiple Choice Questions (10 points)

1. Which prompting technique is most effective for complex reasoning tasks? a) Zero-shot prompting b) Chain of Thought prompting* c) Basic instruction prompting d) Single-shot prompting
- *Correct answer with explanation: Chain of Thought allows for step-by-step reasoning
2. What is the primary purpose of Few-Shot Learning in prompt engineering? a) To reduce token usage b) To provide examples that guide the AI's response format and style* c) To make responses shorter d) To increase processing speed
- *Correct answer with explanation: Few-Shot Learning demonstrates patterns through examples

Short Answer Questions (10 points)

1. PCRO Framework Analysis (4 points) Question: "Explain how the PCRO (Purpose, Context, Requirements, Outcome) framework improves prompt effectiveness."

Rubric: • Purpose clearly explained (1 point) • Context importance described (1 point) • Requirements role detailed (1 point) • Outcome significance explained (1 point)

Scenario-Based Questions (10 points)

Scenario: A law firm needs to analyze 100 contracts for force majeure clauses. Question: Design a prompt using Chain of Thought to assist with this task.

Rubric (5 points):

- Clear initial instruction (1 point)
- Proper reasoning steps (2 points)
- Appropriate output format (1 point)
- Error handling consideration (1 point)

Capstone Prompting Project (40%)

Project Description: Students will design a series of prompts for a complex legal scenario involving multiple stakeholders and requirements.

Rubric:

1. Prompt Design (15 points) • Effective use of advanced techniques (5 points) • Clear structure and progression (5 points) • Appropriate error handling (5 points)
2. Technical Implementation (10 points) • Proper syntax and formatting (3 points) • Efficient token usage (3 points) • Clear output specifications (4 points)
3. Results Analysis (10 points) • Testing methodology (3 points) • Performance evaluation (4 points) • Iteration documentation (3 points)
4. Ethics Consideration (5 points) • Bias mitigation (2 points) • Privacy protection (2 points) • Professional responsibility (1 point)

Implementation

1. Technology Requirements: • Canvas LMS for submission • Access to approved AI platforms (ChatGPT, Claude, etc.) • Basic video recording capabilities • No programming experience required
2. Assessment Conditions: • Individual work • Open resource (including AI tools where specified) • Flexible scheduling to accommodate working professionals • Clear AI attribution requirements
3. Support Provided: • Technical tutorials for AI tools • Examples of proper AI attribution • Templates for documentation • Access to instructor office hours via Zoom

Analysis

Scoring Approach:

- Automated scoring for objective questions
- Rubric-based scoring for subjective components
- Peer review incorporation for collaborative elements

Individual Level Analysis:

- Performance breakdown by technique type
- Strength/weakness identification
- Personalized improvement recommendations
- Progress tracking against learning objectives

Group Level Analysis:

- Class average performance
- Concept mastery distribution
- Common misconceptions identification
- Technique effectiveness comparison

Statistical Reporting:

1. Core Metrics: • Mean scores by assessment component • Standard deviation • Item difficulty index • Discrimination index
2. Advanced Analytics: • Concept correlation analysis • Learning progression tracking • Technique adoption rates • Error pattern analysis
3. Visualization: • Performance distribution charts • Concept mastery heat maps • Progress tracking dashboards

This analysis will inform:

- Individual student feedback and support
- Module content refinement
- Teaching strategy adjustment
- Future assessment design improvements